D7 Review practice

25 AI-style snippets with seeded bugs and a hidden answer key.

D7-L1 Review practice: Python and pandas

Goal first, then line by line: what does each line assume, and which input breaks it?

Review routine for a snippet	Goal first, then each line: what does it assume, and which input breaks it?
Python bugs seeded in this lesson	Missing import, mutable default, utcnow, str = None, truthiness of 0.0, inner join, text max.
When truthiness is fine	Testing a list or dict for emptiness. Not for numbers where 0 is valid.

D7-L2 Review practice: PySpark and Spark SQL

For Spark code, check imports, null behaviour in filters and sums, grain in groupBy and joins, passes over data and MERGE keys.

Spark review checklist	Imports, null behaviour in filters and sums, grain in groupBy, passes over data, MERGE keys, silent renames, NOT IN.
MERGE with repeated source keys	Fails when several source rows match one target row. Deduplicate the source on the merge keys first.
withColumnRenamed on a missing column	A silent no-op. Check df.columns and raise.

D7-L3 Review practice: Databricks SQL and DAX

For SQL, check the dialect, grants, group functions, names and ratios. For DAX, check DIVIDE and the exact filter each ALL removes.

SQL review checklist	Dialect, grants to groups, is_account_group_member, three-level names, ratios from sums, one source row per MERGE key.
DAX review checklist	DIVIDE for every ratio. ALL on the one dimension you mean, never the whole fact table.
Deduplicating a MERGE source in SQL	ROW_NUMBER() OVER (PARTITION BY keys ORDER BY ingestion_timestamp DESC), keep rn = 1.